

Humanvitha Chinnam

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SUMMARY

Software Engineer with 3+ years across FinTech and enterprise banking, building scalable web applications in JavaScript, TypeScript, React, and Next.js, backed by Java and Python on the services and tooling side. Proven track record of shipping polished, user-centric interfaces that drive product adoption and engineering test automation that turns multi-day manual cycles into minutes. Strong foundation spanning frontend development, quality engineering, and complex financial workflows.

TECHNICAL SKILLS

Languages: Python, C, Java, SQL, JavaScript, TypeScript

Frontend: React.js, Next.js, Angular, HTML, CSS, Tailwind CSS, Zustand, Drupal

Backend & APIs: Node.js, Django, GraphQL

Databases & Cloud: MySQL, MongoDB, DynamoDB, Oracle, AWS, GCP, React Query

Automation & Testing: VBA Automation, Python Scripting, TSYs Mainframe Testing, Jest, Unit & Regression Testing

Data & ML: Pandas, NumPy, Matplotlib, Scikit-learn, OpenCV, Tableau, Jupyter Notebook

Tools & Practices: GitHub, GitLab, Gerrit, Jira, Confluence, VS Code, Figma, Postman

EXPERIENCE

Software Engineer

Jul 2025 – Present

Barclays, USA

- Architected and deployed a VBA-based automation platform interfacing directly with the TSYs mainframe for end-to-end credit card change validation — reduced a 2-day manual testing cycle to 45 minutes (~95% time reduction), saving 200+ engineering hours per quarter and accelerating release velocity across the team.
- Engineered a Python-based AI computer vision pipeline leveraging image differencing, OCR, and ML-based pattern recognition to automate validation of customer credit card packet documents — achieved 99.2% defect detection accuracy across text, images, and barcodes, eliminating manual visual review entirely.
- Built an AI-driven rewards verification engine that programmatically validates reward point assignments across 50,000+ transactions per cycle against complex business rules — reduced reward calculation errors by 95% and eliminated error-prone manual spot-checking.
- Designed and developed intelligent TSYs script analysis tools to compare large-scale production scripts impacting customer-facing credit card workflows — improved change visibility by 80%, reduced release-related production incidents by 40%, and minimized risk across thousands of lines of production code.
- Led end-to-end testing strategy for credit card management workflows on the TSYs mainframe covering transaction posting, rewards accrual, statement generation, and UI-layer validation — achieved 99.5% defect escape rate reduction across 12+ production releases while establishing CI/CD-integrated test gates with AI-powered anomaly detection to flag regression risks early.
- Developed predictive test prioritization logic using historical defect data and AI-based risk scoring to optimize test execution order — reduced regression cycle time by 35% while maintaining full coverage, and reduced post-deployment hotfixes.
- Partnered with product, engineering, and compliance teams to translate complex credit card rules and regulatory requirements into structured, auditable automated test scenarios — improved test coverage by 60%, strengthened compliance audit readiness, and ensured correctness across reward, transaction, and fraud-detection flows.

Software Engineer Intern

Jun 2024 – Dec 2024

WalletGyde, Colorado

- Migrated the platform end-to-end from Bubble.io to a production-grade Next.js and DynamoDB application, delivering fully responsive desktop and mobile interfaces that contributed to a 40% increase in user adoption.
- Created a Personal Loan Calculator landing page with real-time estimation logic, directly generating 300+ new customer leads.
- Implemented REST APIs that reduced data retrieval latency by 15%, measurably improving system responsiveness across the platform.
- Researched AI-based matchmaking algorithms, improving successful match rates by 25% over legacy approaches.

Graduate Assistant – Data & Web Systems

Mar 2024 – May 2025

CCPHIT, California State University, Long Beach

- Restructured and standardized dataset management processes to ensure data integrity across research workflows, achieving 99% data accuracy and supporting more reliable academic decision-making.
- Modernized a Drupal-based CMS to streamline updates for course offerings and internship opportunities and automate outreach — resulting in a 25% increase in student enrollments and internship placements.

Front-End Developer

Jul 2021 – Aug 2023

Tata Consultancy Services (TCS), India

- Crafted responsive React.js web interfaces for Lloyds Bank's enterprise banking platform, improving performance and driving a 30% increase in client satisfaction scores
- Automated a regulatory-update workflow on the Lloyds banking platform, eliminating 60% of manual processing time and enabling the client to meet compliance deadlines consistently.

- Authored a reusable React component library adopted across multiple Lloyds product lines, accelerating team development velocity by 25% while maintaining scalability and maintainability.
- Optimized code review and release workflows using Gerrit and GitHub, cutting the review cycle by 40% and shortening project delivery timelines by 30%.
- Integrated RESTful APIs with backend teams to power real-time data flow across customer-facing banking workflows, improving reliability and reducing data-related defects.
- Ensured WCAG 2.1 AA accessibility compliance across Lloyds banking interfaces, implementing semantic markup, keyboard navigation, and screen-reader support to meet UK financial accessibility standards.
- Raised Jest test coverage through component-level testing, reducing production bugs by 40% and improving overall release quality.
- Validated cross-browser and responsive behavior across desktop, tablet, and mobile breakpoints, ensuring consistent rendering and functionality for a broad customer base.
- Improved frontend performance through lazy loading, code splitting, and bundle-size optimization, strengthening Core Web Vitals and reducing initial page load times.
- Contributed actively to Agile ceremonies—sprint planning, stand-ups, and retrospectives—supporting consistent on-time delivery across release cycles.

PROJECTS

HearHub Headphones – E-commerce Platform | *React.js, Sanity, Tailwind CSS, Stripe*

- Launched a full-stack e-commerce platform with Stripe payment processing and a Sanity CMS catalog, boosting user retention by 40% and reducing payment failures by 60%.

Reliever – Social Media Application | *Next.js, TypeScript, GraphQL, Zustand*

- Architected a video-sharing platform with optimized media caching and GraphQL data fetching, cutting load times by 40% and improving search performance by 25%.

Real-Time Chat Application | *React.js, JavaScript, Chat Engine, WebSockets*

- Engineered a live messaging application using WebSockets for bidirectional communication, slashing latency by 50% and increasing user engagement by 30%.

EDUCATION

California State University, Long Beach

Master of Science in Computer Science (GPA: 3.7)

Aug 2023 – May 2025

Lakireddy Bali Reddy College of Engineering

Bachelor of Technology in Computer Science (CGPA: 9.28)

Aug 2017 – Jul 2021

CERTIFICATIONS & PUBLICATIONS

Google Cloud Certified Associate Cloud Engineer

Real-Time Face Mask Detection System

OpenCV, DNN, Transfer Learning (95% accuracy)